CSE 516 – Probabilistic Graphical Models encompasses the major project.

The Deep End-to-End Causal Inference: Bayesian Structure Learning and Simulation-based Treatment Effect Estimation under Heterogeneous, Partially Observed Data

Group G5_TH_PGM: Track Theoretical AI: Repository.

The base implementation is using Microsoft Causica at github.com/masoomchoksi/G5_TH_PGM.

The re-derivation of DECI Theorem 1, a proof-step audit of missing-data boundary, and a prototype of graph-aware imputation were tested in experiment E7.

1. Introduction

Causal questions ask "what would happen if...?Causal questions ask "what would happen if an intervention happens? DECI (Geffner et al., 2022) maintains a distribution over causal graphs, and combines this with learnt structural mechanisms to marginalise graph uncertainty when estimating treatment effects.Our project asks one PGM question: when does likelihood-style learning of a Bayesian network with unknown structure provably recover that structure, and what changes when the model leaves the regime of its assumptions?

M1 conducted the audit for Theorem 1.M2 re-built the proof (M2, Section 8) and the mixed and missing type/date boundaries. M3 does four things: (a) makes a fully observed re-derivation and a missing data proof-step audit; (b) suggests and experiments with an inference side repair targeted at identified boundary.

1.1 Milestone will be used to support the following grade level outcomes:

- A re-derivation of Theorem 1 for the fully observed case, where each assumption is represented by a proof step.
- A proof-step audit that makes it clear what steps are not used as written in the absence of data.
- A The proposed solution is graph-aware imputation.The approach proposed is graph-aware imputation. using the Markov blanket of each missing variable under a sampled DECI graph.
- Controlled rerun of E7 (missingness at 0%, 10%, 30 and 50%), baseline DECI, graph-aware DECI and matched controls. An explicit statement of what can and cannot be claimed. No new Theorem is proven.

1.2 Changes from the M2 report

Item	M2	M3

Missing-data boundary	Proposition 3, to be derived and verified	Audited step by step; Step 1 flagged open, Step 3 identified as the main boundary
New method	None (audit only)	Graph-aware imputation prototype (Section 8)
Experiments	E1, E4–E5 first runs	Only E7 is rerun; other experiments use no missing values and are unchanged
Earlier idea	CI-test / skeleton pruning considered	Replaced by the imputation direction
Numbering	Lemma/Remark labels inconsistent with the paper	Aligned (see Section 6.1)

The scope of the project is otherwise unchanged. Sections of M2 not listed above (graphical representation, learning method, inference method, hyper-parameters) carry over unchanged.

2. Problem formulation

Let x = (x_1,…,x_D) be drawn from a non-linear additive noise SEM with respect to some unknown DAG G^0 whose adjacency matrix is binary :-

$$x_i = f_i\left(x_{\mathrm{pa}(i,G^0)}\right) + z_i,$$

, where z_i is independent of all other variables, i = 1,…,D. For N i.i.d. samples x^1,…,x^N, this report focuses on:

1. When will the optimal conditions of the DECI ELBO be asymptotically equal to (θ, δ(G = G^0))?

2. Where will the arguments fail if we have missing data?Question 1 (validity). Under which conditions do the DECI ELBO optimum conditions converge to (θ*, δ(G = G^0))?

Question 2 (boundary, missing data). If there are missing values, at which proof steps will the argument fall through?

Q3 (Repair from M3). How can we leverage the graphaware posterior from the current graph to reduce degradation from the structural/CAusal effects in comparison to graphagnostic imputations?

Q4 (Empirical). How do our theoretical bounds translate into the performance of graph recovery and

ATE/CATE KPIs?

Contribution. This is a validation and empirical study of the DECI. Our approach is novel as it provides inference-based intervention.

3 Background and Related Work

Background. See M2 Section 3. Two new things are added:

Markov Blanket. Given a node

$$\mathrm{MB}_G(X_j) = \mathrm{Pa}(X_j) \cup \mathrm{Ch}(X_j) \cup \mathrm{Pa}(\mathrm{Ch}(X_j)) \setminus \{X_j\}$$

If the full Markov blanket of $ X_j $ is known, $ X_j $ is conditionally independent of all other variables.

Missing Data Mechanisms. MCAR is assumed in the DECI paper (footnote 3), and this report. MAR and MNAR alter the likelihood of the observed data and are outside of the scope.

Related work and SOTA anchor. See M2 Section 4. As promised in M2, ordering accuracy KPI is also provided as an optional measure for DECI posterior samples. We make no claims of surpassing this.

4. Graphical Representation, Learning and Inference (PGM Mapping)

Vertical	Fully observed DECI	What M3 changes
Representation	Nodes x_i, DAG G, SEM f_i, noise z_i, posterior q_φ(G)	Adds unobserved entries x_u as local latents; Markov blanket of each missing variable read from a sampled graph

Vertical	Fully observed DECI	What M3 changes
Learning	ELBO(θ, φ) with DAG-constrained prior	Missing-data ELBO(θ, φ, ψ) with amortised Gaussian imputer q_ψ(x_u given x_o); prototype adds a graph-aware preprocessing stage and refits
Inference	Posterior over graphs, intervention simulation, ATE/CATE	Unchanged, applied to the model refit on graph-aware completed data

The graph G is a single global latent shared by all N samples. The unobserved entries x_u^n are local latents, one set per sample. This global versus local distinction is the key to the audit in Section 6.

Method for Learning and Inference (Baseline)

The learning goal in the absence of any missing data is

$$ELBO(\theta,\phi) = \mathbb{E}_{q_\phi(G)}\left[\log p(G) + \sum_{n=1}^{N} \log p_\theta(x^n \mid G)\right] + H(q_\phi) \le \log p_\theta(x^1,\ldots,x^N)$$

For missing data (MCAR), the DECI model for p(G, x_u | x_o) factorizes into the amortized product of q_φ(G) and $q_\psi(x_u \mid x_o)$ where x_o^n is the input data. The learning goal becomes

$$ELBO(\theta,\phi,\psi) = H(q_\phi) + \sum_{n=1}^{N} H(q_\psi(x_u^n \mid x_o^n)) + \mathbb{E}_{q_{\phi,\psi}}\left[\log p(G) + \sum_{n=1}^{N} \log p_\theta(x_o^n, x_u^n \mid G)\right]$$

Training and hyper-parameters (λ_s = 5, Gumbel-softmax temperature 0.25, augmented Lagrangian schedule) and ATE/CATE simulation are as in M2 Sections 6, 7 and 9.2. The baseline for E7 is the graph agnostic amortized imputer from above.

6.Mathematical Formulation and Validation

6.1 Re-derivation of Theorem 1 (fully observed case)

Assumptions, as in the paper's Appendix A and M2 Section 8.1:

#	Assumption	Used in
A1	Minimality	Identifiability step
A2	Structural identifiability	Identifiability step
A3	Correct specification	MLE step (bound attained)
A4	Causal sufficiency	MLE step (valid factorisation)
A5	Regularity: $\mathbb{E}_{p^0}[\lvert \log p_\theta(x;G)\rvert] < \infty$	MLE step and LLN

A6– A10	Implicit conditions found in the M2 audit: uniform convergence for exchanging limit and argmax (A6), representability of $\delta(G^0)$ in the variational family (A7), treatment of cyclic graphs (A8), global versus practical optimisation (A9), bounded hyper-parameters (A10)	ELBO step

The chain, using the labels of the original paper (the paper's Lemma 1 is the KL lemma; M2 called it Lemma 2, which is corrected here):

1. Likelihood is a proper density over DAGs. Given that the Jacobian of z = x − f_G(x) on DAGs is triangular with unit determinant, we get
$$p_\theta(x \mid G) = \prod_{i=1}^{D} p_{z_i}\left(x_i - f_i(x_{\mathrm{pa}(i;G)})\right)$$
. 2. Graph KL term will vanish (proof in paper's Lemma 1). As the graph space {0,1}^{D×D} is finite, given bounded λ, ρ, α, KL(q_φ ‖ p) is O(1) uniformly in q_φ, hence (1/N)·KL → 0.
2. MLE recovers the truth (Proposition 1). By Gibbs' inequality,
$$\mathbb{E}_{p^0}[\log p_\theta(X \mid G)] - \mathbb{E}_{p^0}[\log p^0(X)] = -\mathrm{KL}\left(p^0 \parallel p_\theta(\cdot \mid G)\right) \leq 0$$
, with equality iff $p_\theta = p^0$. A3 and A4 make 0 attainable.
3. Identifiability. If A1–A2 make the graph at which equality is achieved equal to G^0, then this is called the graph attaining equality.If A1–A2 make the graph that they are equal to G^0, then it is called the graph attaining equality. 5. ELBO falls on the truth (Theorem 1). Dividing the ELBO by N, step 2 removes the KL term, the LLN (A5, A6) gives the limit
$$\sum_G q(G)\, \mathbb{E}_{p^0}\left[\log p_\theta(x \mid G)\right]$$
, and steps 3–4 give $q' = \delta(G = G^0)$.

This is the result for a fully observed continuous additive-noise model. Properly, the KL bound and pointwise LLN alone do not establish convergence of the optimiser; A6–A9 are needed and are reported as gaps in M2 Section 8.3.

6.2 Proof-step audit under missing data

Proof step	Fully observed	With missing data	Status

1. MLE recovers truth	Complete-data likelihood and identifiability	Becomes an observed data likelihood problem. Identifiability from observed entries needs a new lemma, which this report does not supply	**Changed; open. Not addressed by our method**
2. Graph KL term → 0	Finite graph space, bounded prior	Graph is still a single global latent; this step is not the failure identified by the audit	**Unchanged in the audit**
3. ELBO collapses onto the true graph	Bound can meet the ideal objective	An imputation gap can stop the bound from reaching the complete data optimum	**Changed; main target**

6.3 The imputation gap (Proposition 3, restated)

Under MCAR the missing-data objective satisfies

$$ELBO(\theta, \phi, \psi) = \log p_\theta(x_o^{1:N}) - \mathrm{KL}\left(q_\phi(G)\prod_{n=1}^{N} q_\psi(x_u^n \mid x_o^n) \parallel p_\theta(G, x_u^{1:N} \mid x_o^{1:N})\right)$$

If the posterior graph has shrunk to $\delta(G^0)$, then the gap is

$$\sum_{n=1}^{N} \mathrm{KL}\left(q_\psi(x_u^n \mid x_o^n) \parallel p_\theta(x_u^n \mid x_o^n, G^0)\right)$$

These terms are all non-negative, and zero when the Gaussian family is equal to the actual conditional. For a non-linear SEM with non-Gaussian noise, the real conditional is typically not Gaussian (i.e. skewed or multimodal) and each term can be bounded away from 0 and the sum is $\Theta(n)$, while KL is $O(1)$. Therefore (1/N) × gap is not going to be zero, and (1/N) × ELBO maximiser $\hat{\theta}$ is not necessarily unbiased relative to the MLE as N becomes large. In special cases the gap is zero, like in the case of a linear-Gaussian SEM that we use as a control.

The two sources of the gap. A family mismatch occurs if a Gaussian q_ψ is not matching a non Gaussian conditional (i). Graph-agnosticism: Conditioning on x_o based on q_ψ without invoking G: Only source (ii) is addressed with graph-aware conditioning.

6.4 Wording of claims

We say	We do not say
Missing data is outside the fully observed setting of the proof.	We relaxed one of Theorem 1's five assumptions.
The proof no longer applies as written.	Theorem 1 is false under missing data.
Our method targets one component of the imputation gap.	Our method proves the theorem under missing data.
We test whether graph-aware imputation improves empirical recovery.	Graph-aware imputation is guaranteed to improve recovery.

7. Theorem 1 Proof-Gap Assessment and Gaps Addressed in M3

7.1 Gaps in the DECI paper relevant to M3

#	Gap	Source	Handled in M3
G1	Theorem 1 does not cover missing data; the paper defers a guarantee to future work	Paper, Appendix A	Audit (Section 6.2); E7
G2	Missing data handled only by a Gaussian, graph-agnostic amortised imputer; the imputation gap is nonzero for non-linear, non-Gaussian SEMs	Paper Sec. 3.4; M2 Prop. 3	Graph-aware imputer (Section 8)

G3	Missing-data evidence is thin: one 30% setting, graph metrics only, no ATE/CATE under missingness, no sweep, no imputation error	Paper Figs. 4 and 7	E7 at 0/10/30/50% with imputation, graph and causal KPIs

#	Gap	Source	Handled in M3
G4	Only MCAR is considered	Paper footnote 3	Stated limitation; MAR as optional extension
G5	Lemma 1 assumes bounded λ, ρ, α, but training raises ρ and α towards 10^{13}	PaperAppendix B.1	Flagged (A10); derivation work, no new compute
G6	No uniform-convergence argument for exchanging limit and argmax	PaperAppendix A.3	Flagged (A6)
G7	Seeds and statistics: 5 seeds for ER/SF, no confidence intervals or paired tests	Paper Tables 3–5	Paired seeds, bootstrap CIs, effect sizes
G8	Posterior calibration of q_φ never evaluated	Paper Sec. 4	Optional edge Brier / ECE under missingness

7.2 Assessment of the three proof steps

Step	Label	Assessment

1. MLE recovery	**Open; not targeted by the method**	Better imputations may help recover the underlying distribution empirically, but identifiability from observed entries remains unproved. We flag it as future work and, at most, a conjecture under MCAR
2. Graph KL term	**Not the primary gap**	No change to this proof component is proposed
3. ELBO convergence	**Directly targeted (empirically)**	Graph-aware imputation explores the graph-agnostic component of variational approximation gap. It does not make mathematically the gap disappear and the prototype does not change the ELBO.

The strongest gap for the milestone is Step 3, which is supported by the scope statement of the paper, it has a concrete mechanism (Θ(N) versus O(1)), and it maps directly onto experiment E7.

8. Proposed Solution: Graph-Aware Imputation

8.1 Idea

The imputer replaces an undifferentiated conditioning set with the local graphical neighbourhood of each missing variable, thus using a sampled graph G.For a missing X_j it estimates

$$q_G\left(X_j^u \mid \mathrm{MB}_G(X_j),\, M\right)$$

where M stands for the missingness mechanism. The imputation procedure depends on G, that is, the current graph hypothesis determines the conditioning neighbourhood. If blanket members are missing, the imputation procedure is repeated.

Example. With edges X1→X3, X3→X4, X2→X4, the blanket of X3 is {X1, X4, X2}.

8.2 How it can support or not support

The set of variables is a subset of the variables that the amortiser sees already. In principle, the method does not add any information. It would be nice to obtain any benefit from inductive bias and sample efficiency: a lower dimensional, properly targeted conditioning set. Graph errors can propagate into the imputations if the graph is not right.Measuring this failure mode is part of the evaluation.

8.3 Imputer variants

Variant	Description	Role
A. Baseline DECI	Original graph-agnostic amortised imputation	Required control
B. Mean/median fill	Simple initialisation, then refit	Floor
C. Agnostic MLP	Same-capacity MLP using all other variables, not the blanket	**Mandatory control**: separates graph structure from imputer capacity

Variant	Description	Role
D. Graph-aware, deterministic	MLP predicts the missing value from blanket variables	First working prototype
E. Graph-aware, stochastic	MLP outputs distribution parameters and samples	Preferred main method; also addresses family mismatch partly
F. Oracle graph	Variant D or E with the true DAG	Diagnostic only, never mixed into the fair comparison

A deterministic MLP gives conditional means and is not a valid conditional sampler, so variant E is the more defensible method. Neither variant makes the imputation gap zero.

8.4 Handling missing blanket members in an iterative manner

1. Initialize each missing value (median, mean or current DECI imputation). 2. For each missing target variable X_j, find its blanket variables.

3. Apply the current and previous imputed values of the blanket variables that are missing. 4. Impute a new value for X_j.

5. Repeat for K times; check the average change in the imputed value at a later stage to determine convergence.

8.5 Two-stage prototype and its limitations

The prototype preprocesses the data and re-trains DECI, and thus does not affect the ELBO. The prototype can be seen as an empirical approximation of the gap, but not as its reduction within the objective. The incorporation of the variational imputer within the ELBO is a more advanced research direction than M3. The approach addresses only the graph-oblivious part of the gap; the gap caused by Gaussian family mismatch and Step 1 identifiability is still open. MCAR assumption is made throughout all analyses.

9. Implementation

9.1 Pipeline

Graph aware conditional imputation: Generate synthetic dataset with known DAG structure (0/10/30/50% of data missing) → Run DECI on the incomplete dataset → Graph samples from posterior distribution → Markov blankets of variables with missing values → Graph aware conditional imputation iteratively → Observed values used to get the completed dataset → DECI re-training with the same budget and random seeds → Evaluation of the graph, imputation and causal metrics (against the observed data).

Same mask used for all methods for each seed and rate. Observed entries are never overwritten. Graph extraction must use the API of the installed Causica version, not a name assumed from another release.

9.2 Code organisation

File	Purpose
data_missingness.py	Create and save masks (complete, mask, incomplete data)
markov_blanket.py	Graph parsing and blanket construction
graph_aware_imputer.py	Conditional model and iterative imputation
graph_sampling.py	Wrapper over the installed DECI graph-posterior API
run_baseline_deci.py	Unchanged baseline entry point
run_graph_aware_deci.py	Experimental wrapper
metrics.py	Graph and imputation metrics

run_e7.py, results.py	Missingness × seed loop; aggregation, CIs and plots

9.3 Unit tests and debugging sequence

Tests: blanket test for hand written 4 node DAG; no self-node in blanket; duplicates eliminated; 0% missingness is equal to data; observed entries not altered; a totally missing target variable is processed; missing blanket variables by iterating; invalid graph structure rejected; imputation error calculated from hidden entries only.

The debugging order is masks only; blankets only;imputer with a fixed, known graph; very small synthetic iterations; an oracle graph; one sample of a DECI graph:A complete comparison at 0%, then at 10%, 30% and 50%.

9.4 Status of Results

Stage	Status
Baseline DECI (M2 pipeline)	Code written and executed in M2
Missingness masks, blanket and imputer modules	To be completed and tested before the E7 runs
E7 runs and statistics	Pending; no E7 results are reported in this document

10. Experimental Setup (E7 only)

Only E7 will be rerun. E1-E6 and E8-E10 have no missing data and will not use the new imputer. Their M2 results will not change.

Factor	Levels / control	Primary comparison
Missingness (MCAR)	0%, 10%, 30%, 50%	Baseline DECI vs graph-aware DECI vs controls
Seeds	At least 5 per cell, fixed and listed in the config	Mean ± SD and 95% CI

Graph and data	Same DAG and data generation for paired methods (start with ER(16,16))	All non-missingness factors controlled
Model budget	Same epochs, batch size, architecture, hardware	No extra compute for one method
Ground truth	Known DAG; known effects where feasible	SHD/F1; ATE/CATE error

0% sanity check. Every experiment must match DECI baseline at 0% missingness. Otherwise the implementation is reviewed before any missingness analysis is considered.

10.1 Key Performance Indicators

PGM stage	KPI	Direction
Representation	SHD, adjacency F1, orientation F1	↓ / ↑
Imputation (diagnostic)	RMSE and MAE on artificially hidden entries only	↓
Imputation gap proxy	Held-out NLL; on synthetic data, KL to the known conditional where computable	↓
Learning	ELBO, held-out NLL	↑ / ↓
Inference	ATE / CATE RMSE, if the pipeline supports them reliably	↓
Uncertainty (optional)	Edge Brier / ECE, ordering accuracy	↓ / interpret
Cost	Runtime split into stage 1, imputation and refit; imputation sweeps	↓

Imputation RMSE/MAE are diagnostics and cannot be used to replace graph recovery metrics.

10.2 Ablations

Equal capacity agnostic MLP vs graph-aware (graph structure vs imputer capacity)

One versus several (for example 5) posterior graph samples.

Estimated graph versus oracle graph (diagnostic only).

Optional robustness: noise type, graph size, and a linear-Gaussian control where the gap is expected to be zero.

10.3 Hypothesis

H1. Graph-aware imputation avoids degradation in the recovery of the graph and estimation of causal effects due to missingness, particularly at moderate to high rates of missingness and when the stage-1 graph is informative. The two methods should be in agreement at 0% and the difference between them might be very small at low rates, however the imputer can propagate errors with a poor graph.

10.4 Statistical Analysis Plan

Mean ± SD per cell, 95% confidence interval (using t-statistic if the sample size is small; otherwise bootstrap with a sample size of 2000).

Paired Tests. For each seed/mask combination, we will perform a t-test/bootstrap confidence interval or Wilcoxon signed rank test with enough pairs (around 6). We will compute the effect size using Cohen's d_z.

Significance is never used as proof of a theorem; it is read alongside which assumptions hol

11. Results (E7)

The supplied E7 screenshots report DECI-G baseline results for the linear-Gaussian and nonlinear regimes at 0%, 10%, 30% and 50% missingness. Values below are transcribed as mean ± standard deviation from the result tables (λ = 5). Imputation RMSE is evaluated on hidden entries; the 0% rows have no imputation error because no entries are missing.

11.1 E7 summary — DECI-G baseline

Regime	Missing	Expected SHD ↓	Adj. F1 ↑	Orient. F1 ↑	Imputation RMSE ↓	Mean-fill RMSE ↓	Runtime (h)
Linear-Gaussian	0%	53.03 ± 2.76	0.337 ± 0.014	0.089 ± 0.029	—	—	0.48 ± 0.04
Linear-Gaussian	10%	42.17 ± 7.14	0.449 ± 0.090	0.215 ± 0.060	0.643 ± 0.061	1.011 ± 0.013	0.66 ± 0.08
Linear-Gaussian	30%	42.46 ± 6.49	0.442 ± 0.049	0.207 ± 0.063	0.657 ± 0.049	0.992 ± 0.019	0.52 ± 0.05
Linear-Gaussian	50%	41.21 ± 3.78	0.474 ± 0.035	0.180 ± 0.059	0.725 ± 0.046	1.003 ± 0.010	0.43 ± 0.04

Nonlinear	0%	40.26 ± 3.01	0.468 ± 0.011	0.216 ± 0.031	—	—	0.54 ± 0.02
Nonlinear	10%	47.36 ± 3.88	0.380 ± 0.026	0.126 ± 0.024	0.807 ± 0.011	1.000 ± 0.003	0.60 ± 0.04
Nonlinear	30%	42.10 ± 3.93	0.429 ± 0.031	0.197 ± 0.014	0.816 ± 0.046	1.001 ± 0.013	0.46 ± 0.06
Nonlinear	50%	36.87 ± 4.54	0.484 ± 0.066	0.237 ± 0.063	0.855 ± 0.036	0.995 ± 0.010	0.43 ± 0.03

ATE/CATE RMSE is not present in the supplied E7 outputs, so it is not reported here. The screenshots also do not provide results for graph-aware imputation, agnostic MLP, or oracle-graph variants; those cells must remain unreported until the corresponding runs are completed.

11.2 Interpretation of the available baseline results

Linear-Gaussian regime. Across the reported missingness levels, expected SHD ranges from 41.21 to 53.03. Adjacency F1 is highest at 50% missingness (0.474 ± 0.035), while orientation F1 is highest at 10% (0.215 ± 0.060). Imputation RMSE increases from 0.643 ± 0.061 at 10% to 0.725 ± 0.046 at 50%, indicating greater reconstruction error as the proportion of hidden entries increases.

Nonlinear regime. At 10% missingness, adjacency F1 decreases from the 0% reference value of 0.468 ± 0.011 to 0.380 ± 0.026, and orientation F1 decreases from 0.216 ± 0.031 to 0.126 ± 0.024. At 30% and 50%, the reported adjacency F1 values are 0.429 ± 0.031 and 0.484 ± 0.066, respectively. Imputation RMSE rises from 0.807 ± 0.011 at 10% to 0.855 ± 0.036 at 50%. These results are descriptive; the non-monotonic graph metrics do not establish that higher missingness improves graph recovery.

Runtime. The reported baseline runtimes range from 0.43 to 0.66 hours in the linear-Gaussian regime and from 0.43 to 0.60 hours in the nonlinear regime. Runtime alone does not establish a performance advantage for any imputation strategy.

11.3 Additional DECI validation results shown in the screenshots

DECI-G main arm ($\lambda = 5$, n = 1000; mean ± SD over 3 seeds). The separate main-arm table reports the following graph-recovery results:

D (edges)	Expected SHD ↓	Adj. F1 ↑	Cyclic-sample rate ↓	Shortest cycle (median)	Runtime (h)
8 (8)	10.91 ± 1.97	0.572 ± 0.062	0.000 ± 0.000	none cyclic	0.57 ± 0.19
12 (12)	25.25 ± 4.96	0.498 ± 0.111	0.000 ± 0.000	none cyclic	0.46 ± 0.07
16 (16)	40.26 ± 3.01	0.468 ± 0.011	0.000 ± 0.000	none cyclic	0.59 ± 0.06
20 (20)	61.65 ± 7.93	0.365 ± 0.035	0.000 ± 0.000	none cyclic	0.59 ± 0.05

24 (24)	81.30 ± 5.81	0.365 ± 0.026	0.000 ± 0.000	none cyclic	0.75 ± 0.03
32 (32)	119.89 ± 0.45	0.319 ± 0.010	0.000 ± 0.000	none cyclic	0.75 ± 0.04

DECI-S validation arm (λ = 5, n = 1000; mean ± SD over 2 seeds). For D = 8 (8), expected SHD is 9.67 ± 2.09, adjacency F1 is 0.663 ± 0.082, cyclic-sample rate is 0.005 ± 0.007, median shortest cycle is 3, and runtime is 1.58 ± 0.02 hours. For D = 24 (24), expected SHD is 69.78 ± 0.77, adjacency F1 is 0.365 ± 0.039, cyclic-sample rate is 0.000 ± 0.000, no cyclic sample is reported, and runtime is 4.84 ± 0.70 hours.

11.4 E7 run-log status and limitations

The terminal screenshots show DECI-G baseline run records for the linear-Gaussian and nonlinear regimes. In the displayed log, the 10% cells have 3/3 runs for both regimes; the nonlinear 30% cell is shown with 1/1 run, while the linear-Gaussian 30% cell is marked “Not run” in that log snapshot. The summary tables supplied separately report aggregated values for 0%, 10%, 30% and 50%; the report therefore preserves those table values but flags that the accompanying terminal snapshot does not independently demonstrate complete run coverage for every cell. Verify the raw per-run CSV before final submission.

The supplied screenshots do not show E7 outputs for graph-aware DECI, equal-capacity agnostic MLP, oracle graph, ATE/CATE RMSE, confidence intervals, paired tests, or effect sizes. No comparison or causal-effect conclusion about these methods can be made from the available evidence. The graph-aware prototype is a two-stage preprocessing-and-refitting procedure and does not modify the DECI ELBO; this limitation remains as stated in Sections 8.5 and 13 of the report.

11.5 Summary of supported findings

The available results establish a baseline reference for DECI-G under the two displayed data regimes. Imputation error generally increases with missingness in both regimes, while graph-recovery metrics vary non-monotonically. The DECI-G main-arm table also shows that expected SHD increases and adjacency F1 generally decreases as the number of graph edges increases. These observations are descriptive and do not test the central M3 hypothesis that graph-aware imputation improves recovery relative to graph-agnostic controls.

Required before claiming the full E7 comparison: complete and verify the missing graph-aware, agnostic-MLP and oracle runs; report ATE/CATE only if reliably produced; aggregate all paired seeds; and calculate the planned confidence intervals and paired comparisons. Until then, the Section 11.1 table should be described as the available DECI-G baseline summary, not the completed multi-method E7 comparison.

11.6 Intended graphs

Missing vs SHD & F1; missingness vs imputation error; run time; stage 1 SHD vs imputer gain (error analysis); edge reliability plot (optional).

11.7 Summary Template for Major Results

What happened with this major result? Was it expected from H1? Why? (local conditioning based on

the causal graph, versus imputer capability.) How does it affect the frontier of missing data? Does H1 hold? Limitation: propagation of graph errors, single imputation treated as real data, only MCAR.

11.8 How outcomes will be read

Outcome	Interpretation
Lower SHD/F1 degradation, similar runtime	Strong empirical support for the proposed repair
Better graph metrics, worse imputation error	Local conditioning may help graph learning despite imperfect value reconstruction
Better imputation, no graph improvement	The imputer works locally but does not repair graph learning
No meaningful difference	Graph information not useful at tested rates, or the prototype is too weak
Agnostic MLP matches graph-aware	Benefit comes from imputer capacity, not graph structure
Worse results	Important failure case; graph errors amplified

11.9 Carried over from M2 (unchanged)

The E1 and E4–E5 results and interpretation are given in M2 Section 11 and are not repeated. Two corrections for the final submission: The caption for the E1 table must contain the same number of seeds as the text, and the gap between the DECI and the PC/NOTEARS-MLP should be checked for convergence and cyclic-sample rate and hyper-parameters before concluding that P1 is only partially supported.

12. Claims We Can and Cannot Make

Can claim if supported by results	Cannot claim without a new proof
The graph-aware imputer improves or worsens E7 performance under the tested conditions	Theorem 1 now holds under missing data

The method reduces imputation error at X% missingness	The imputation gap is mathematically zero
Graph quality affects the usefulness of graph-aware imputation	Graph-aware imputation guarantees consistency
The method is an empirical repair targeting the identified proof boundary	We have extended the original theorem

13. Discussion and Limitations

The guarantee is a chain with three links: the Gibbs' inequality (distribution), identifiability (distribution to graph) and a counting argument (one global latent variable ensures the variational penalty is low). In the case of categorical variables, the second link falls apart; for amortized imputation, the third link fails; the first link holds in both cases.

Limitations.

Open proof step. The identifiability of the observed data (Step 1) is not proven here. Partial solution. The method works on the graph-agnostic aspect of the question but not on the Gaussian family mis-specification aspect.

Two-stage method. Graph comes from a baseline trained on badly imputed data and is at its worst precisely where the effect is supposed to occur (30-50%); refitting on a single imputation means treating imputed values as observed data without considering imputation uncertainty. Multiple graphs or multiple imputations helps address this problem. Indirect metrics. Only indirectly do SHD and F1 measure the imputation gap; the held-out NLL and KL metric proxy help connect it to the audit.

MCAR only. MAR and MNAR alter the observed data likelihood and therefore the argument of identifiability.

Optimization. The asymptotic result applies to global optimization but the finite sample and optimizer (A9) behavior may differ.

Mean field posterior. q_φ can't capture edge dependencies, thus, probability of G^0 under the posterior is an attribute of q_φ and not of the actual posterior.

Propositions 2 and 3 are propositions of our audit, formulated precisely but not yet peer-reviewed.

Computational details. Limited to seeds and sizes of graphs, missing cells are printed not silently ignored.

14. Reproducibility

- Please use the following directory structure for repos: /theory (proofs), /src (generator, imputer,

metrics, stats), /configs (one config per experiment, seeds), /results (raw CSV per run), /plots, /docs. Please save everything: data, masks, incomplete data for each seed and rate; masks are reproducible and are shared across the methods.

- Record Python, PyTorch, Causica and gcastle versions from the final environment log in the README; do not infer them.
- One script per table, e.g. python run_e7.py --config configs/e7.yaml . No undocumented local files, preparation steps for documentation of data sets.
- Utilization of generative AI: AI tools can be used to generate content, brainstorming and coding assistance, and each member is required to be familiar with the content and to declare its use in the individual form.

Last minute checklist before the E7 run

Baseline DECI runs at 0% missingness and masks are stored.

Universal function passes manual DAG test checks; graph extraction works with the installed version of Causica.

Imputer does not overwrite observed values; Oracle works before the graph-estimation version.

Solution for 0% graph-dependent is close to the baseline; all inputs are identical..

15. Team Responsibilities

Workstream	Assigned member	M3 responsibility
Theory / proof	Parisha Kathiria	Fully observed re-derivation; proof-step audit write-up
Gap analysis	Vrunda Patel	Missing-data audit, imputation-gap derivation, claims section
DECI implementation	Yashvi Patel	Causica wrapper, graph-posterior extraction, refit pipeline
Baselines	Masoom Choksi	Baseline and control variants, same masks and seeds
Datasets &	Khushi Thakar	Mask generation, E7 runs across rates and seeds

experiments		
Evaluation & visualisation	All members	KPIs, CIs, paired tests, plots, interpretation

16. Conclusion

M3 brings closure to the fully observed re-derivation, checks the missing steps of data (Step 1 open, Step 2 same, Step 3 the major boundary) and proposes an imputation prototype based on the graph that requires testing only in E7. Contribution is an honest and bounded contribution, which is an empirical examination of one aspect of the imputation gap with clear controls but no new theorem. Next up would be completing the modules, running E7 over paired seeds, and writing the analysis of results.

17. References

1. T. Geffner, J. Antoran, A. Foster, W. Gong, C. Ma, E. Kiciman, A. Sharma, A. Lamb, M. Kukla, N. Pawlowski, M. Allamanis, C. Zhang, "Deep End-to-end Causal Inference," arXiv:2202.02195v2, 2022.

A. Vashishtha, A. G. Reddy, A. Kumar, S. Bachu, V. N. Balasubramanian, A. Sharma, "Causal Order: The Key to Leveraging Imperfect Experts in Causal Inference," ICLR, 2025.

3. J. Peters, J. M. Mooij, D. Janzing, B. Schölkopf, "Causal Discovery with Continuous Additive Noise Models," JMLR, 2014.

4. P. O. Hoyer, D. Janzing, J. M. Mooij, J. Peters, B. Schölkopf, "Nonlinear Causal Discovery with Additive Noise Models," NIPS, 2008.

A. Hyvärinen, A. Kerminen, M. Jordan, P. O. Hoyer and 5. S. Shimizu, "A Linear Non-Gaussian Acyclic Model for Causal Discovery," JMLR, 7, pp. 210-220, 2006.

6. X. Zheng, B. Aragam, P. Ravikumar, E. P. Xing, "DAGs with NO TEARS," NeurIPS, 2018. 7. C. Dan, B. Aragam, X. Zheng, P. Ravikumar, E. P. Xing, "Learning Sparse Nonparametric DAGs", AISTATS, 2020.

8. S. Lachapelle, P. Brouillard, T. Deleu, S. Lacoste-Julien, “Gradient-Based Neural DAG Learning”, arXiv:1906.02226, 2019.

9. I. Khemakhem, R. Monti, R. Leech, A. Hyvärinen, Causal Autoregressive Flows, AISTATS, 2021.

10. D. B. Rubin, "Inference and Missing Data," Biometrika, 63(3), 1976.

Causica, "Causica" by Microsoft Research, GitHub repository, 2022.

https://github.com/microsoft/causica

12. K. Zhang et al., "gCastle: A Python Toolbox for Causal Discovery," arXiv:2111.15155, 2021. 13. M3 - Re-derivation, Missing Data and Experiments.